



INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Energy Consumption Analysis

About me

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Objective Of The Analysis

The primary objective of this analysis is to gain deep insights into global energy usage, production, emissions, Population growth and economic performance(GDP) by country and year. The key goals include:

1. Analysing the relationship between **energy production/consumption and emissions**.
2. Identifying **top contributors to emissions** and **trends over time**.
3. Evaluating **GDP and population growth impacts** on energy demand and emissions.
4. Comparing **per capita metrics** and identifying **major economies energy behaviour**.
5. Measuring **the efficiency of energy usage relative to economic output**.
6. Highlighting global emission shares and identifying countries with positive or negative trends.
7. Supporting sustainable energy policy recommendations based on data-driven findings.

About Tables

Emission Table

energy type	year	emission
Consumed natural gas (MMtonnes CO2)	2020	0
Consumed natural gas (MMtonnes CO2)	2020	0
Consumed natural gas (MMtonnes CO2)	2020	0
Consumed natural gas (MMtonnes CO2)	2020	0
Consumed natural gas (MMtonnes CO2)	2020	0
Consumed natural gas (MMtonnes CO2)	2020	0
Consumed natural gas (MMtonnes CO2)	2020	0
Consumed natural gas (MMtonnes CO2)	2020	0
Consumed natural gas (MMtonnes CO2)	2020	0
Consumed natural gas (MMtonnes CO2)	2020	0

Country Table

Country	CID
Country	CID
Afghanistan	Af0
Albania	Al1
Algeria	Al2
American Samoa	Am3
Angola	An4
Antarctica	An5
Antigua and Barbuda	An6
Argentina	Ar7
Armenia	Ar8

Population Table

Countries	year	value
Afghanistan	2024	42647.49
Albania	2024	2791.765
Algeria	2024	46814.31
Angola	2024	37885.85
Antigua and Barbuda	2024	93.772
Argentina	2024	45696.16
Armenia	2024	2973.84
Aruba	2024	108.066
Australia	2024	27236.21
Austria	2024	9170.854

Production Table

country	energy	year	production
Afghanistan	Nuclear (quad Btu)	2020	0
Albania	Nuclear (quad Btu)	2020	0
Algeria	Nuclear (quad Btu)	2020	0
American Samoa	Nuclear (quad Btu)	2020	0
Angola	Nuclear (quad Btu)	2020	0
Antarctica	Nuclear (quad Btu)	2020	0
Antigua and Barbuda	Nuclear (quad Btu)	2020	0
Aruba	Nuclear (quad Btu)	2020	0
Australia	Nuclear (quad Btu)	2020	0

Consumption Table

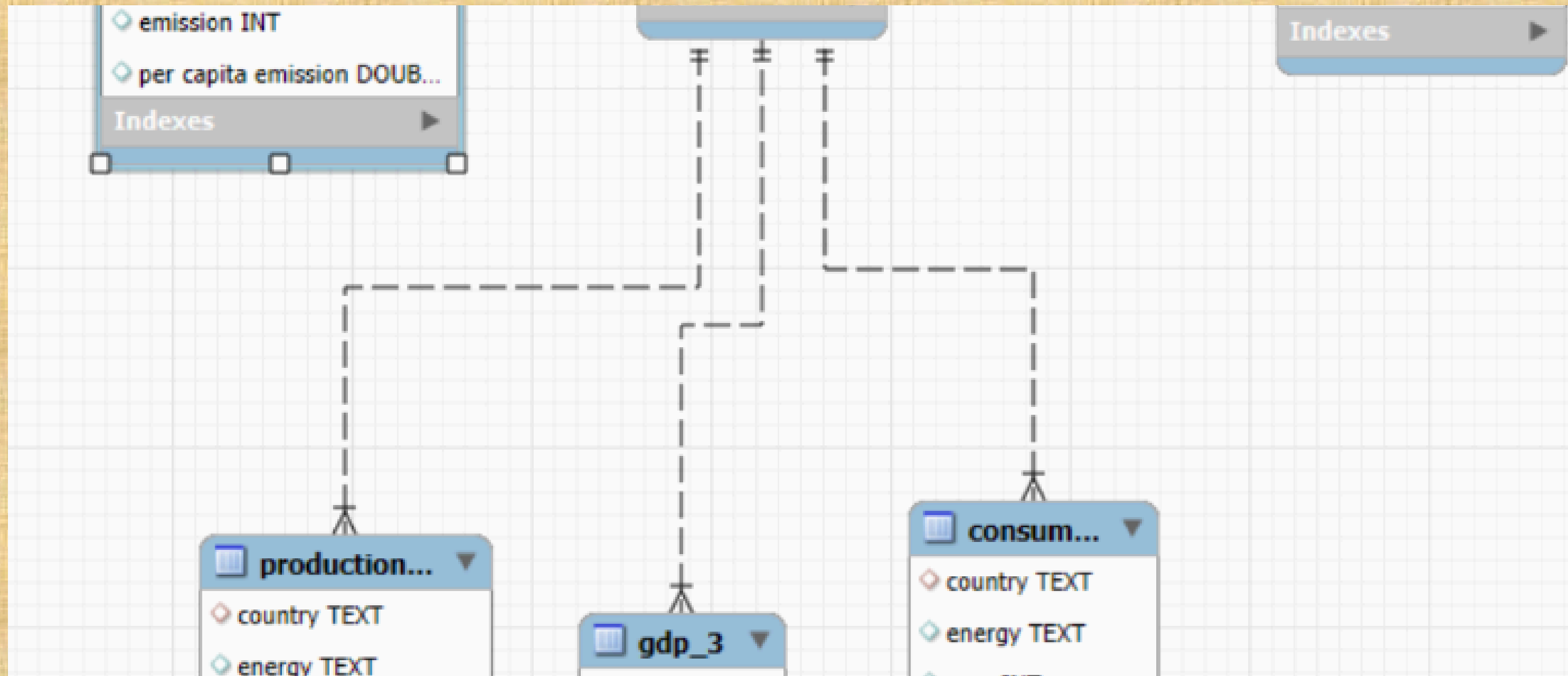
country	energy	year	consumption
Afghanistan	Nuclear (quad Btu)	2020	0
Albania	Nuclear (quad Btu)	2020	0
Algeria	Nuclear (quad Btu)	2020	0
American Samoa	Nuclear (quad Btu)	2020	0
Angola	Nuclear (quad Btu)	2020	0
Antarctica	Nuclear (quad Btu)	2020	0
Antigua and Barbuda	Nuclear (quad Btu)	2020	0
Aruba	Nuclear (quad Btu)	2020	0
Australia	Nuclear (quad Btu)	2020	0
Austria	Nuclear (quad Btu)	2020	0

GDP Table

Country	year	value
Afghanistan	2020	83.21645
Albania	2020	36.78752
Algeria	2020	531.9749
Angola	2020	215.9016
Antigua and Barbuda	2020	1.772876
Argentina	2020	866.9691
Armenia	2020	33.00228
Aruba	2020	2.584789
Australia	2020	1210.203

ER-Diagram

- All the tables are connected with country table.
- One to many relationship exists between them.
- Country is parent table and remaining are child table



1.What is the total emission per country for the most recent year available?

Query

```
SELECT country, sum(emission) as total_emission_3
FROM emission_3
WHERE year=
(SELECT max(year) as recent_year
FROM emission_3)
GROUP BY country
ORDER BY total_emission desc;
```

Output

China	24392
United States	9590
India	5642
Russia	3688
Japan	1920
Indonesia	1659
Iran	1646
Saudi Arabia	1313
South Korea	1288
Germany	1200

Top 3 co2 emitter countries are China, USA and India.

2.What are the top 5 countries by GDP in the most recent year?

Query

```
country, value, year
3
r = (SELECT MAX(year) FROM
value DESC
```

Output

country	value	year
	28673.24	20
United States	22679.47	20
	11660.21	20
n	5179.704	20
any	4463.949	20

3. Compare energy production and consumption by country and year.

Query

```
p.country,  
p.year,  
p.energy,  
c.consumption,  
p.production  
FROM production_3 AS p  
JOIN consum_3      AS c  
  ON p.country = c.country  
  AND p.year    = c.year  
  AND p.energy  = c.energy  
ORDER BY p.production DESC;
```

Output

	year	energy	consumption	
	2023	Coal (quad Btu)	100	9
	2022	Coal (quad Btu)	96	9
	2021	Coal (quad Btu)	95	9
	2020	Coal (quad Btu)	92	9
tes	2023	Natural gas (quad Btu)	34	3
tes	2022	Natural gas (quad Btu)	33	3
tes	2021	Natural gas (quad Btu)	32	3
tes	2020	Natural gas (quad Btu)	32	3
tes	2023	Petroleum and other liquids (quad Btu)	37	3
tes	2022	Petroleum and other liquids (quad Btu)	37	3
tes	2021	Petroleum and other liquids (quad Btu)	37	3

4.Which energy types contribute most to emissions across all countries?

Query

```
, sum(emission) AS tot
_3
rgy type`
1_emission DESC;
```

Output

	to
MMtonnes CO2)	142
MMtonnes CO2)	639
her liquids (MMtonnes CO2)	472
al gas (MMtonnes CO2)	314

5. How have global emissions changed year over year?

Query

```
SELECT  
    sum(emission) AS total_emission  
FROM emissions  
GROUP BY year  
ORDER BY year DESC;
```

Output

year	total_emission
2023	74161
2022	72445
2021	70976
2020	67852

As we see total emission increasing over the year

6.What is the trend in GDP for each country over the given years?

Query

```
SELECT country, year, value AS gdp
FROM gdp_desc,year;
```

Output

country	year	gdp
China	2024	28673.24
China	2023	27313.92
China	2022	25910.6
China	2021	25123.27
China	2020	23136.77
United States	2024	22679.47
United States	2023	22062.56
United States	2022	21443.41
United States	2021	20917.86
United States	2020	19723.57

7.How has population growth affected total emissions in each country?

Query

```
SELECT
p.countries,
p.year,
sum(e.emission) AS total_emission,
p.value AS population
FROM population_3 p
JOIN emission_3 e
ON p.countries = e.country
AND p.year = e.year
GROUP BY p.countries, p.year, p.value
ORDER BY countries,year;
```

Output

countries	year	total_emission	population
Afghanistan	2020	18	39068.98
Afghanistan	2021	20	40000.41
Afghanistan	2022	18	40578.84
Afghanistan	2023	16	41454.76
Albania	2020	6	2871.954
Albania	2021	8	2849.635
Albania	2022	9	2827.608
Albania	2023	7	2811.655
Algeria	2020	284	44042.09
Algeria	2021	304	44761.1
Algeria	2022	317	45477.38
Algeria	2023	328	46164.22
Angola	2020	29	33451.13

8.Has energy consumption increased or decreased over the years for major economies?

Query

```
c.year,  
SUM(c.consumption) AS total_consumption  
FROM consum_3 AS c  
JOIN (SELECT  
country,SUM(value) AS total_gdp  
FROM gdp_3  
GROUP BY country  
ORDER BY total_gdp DESC  
LIMIT 5  
) AS me  
ON c.country = me.country  
GROUP BY c.year,me.country
```

Output

country	year	total_consumption
China	2023	177
Germany	2023	11
India	2023	38
Japan	2023	19
United States	2023	111
China	2022	168
Germany	2022	11
India	2022	36
Japan	2022	19
United States	2022	112
China	2021	165
Germany	2021	13
India	2021	33

Indias energy consumption across year 2021-33,2022-36,2023-38

9. What is the average yearly change in emissions per capita for each country?

Query

```
capita emission',  
OVER (PARTITION BY country ORDER BY year) A  
  
capita emission' - pre_year_emission), 2)  
  
T NULL  
  
change DESC;
```

Output

country	avg_year_percapita_change
Afghanistan	0
Albania	0
Algeria	0
American Samoa	0
Angola	0
Antarctica	0
Antigua and Barbuda	0
Argentina	0
Armenia	0
Aruba	0
Australia	0
Austria	0
Azerbaijan	0

10. What is the emission-to-GDP ratio for each country by year?

Query

```
ry, e.year,  
mission)/sum(g.value)), 4) AS emis  
3 e  
  
g.country  
.year  
ry, year  
ry,year;
```

Output

country	year	emission_gdp_ratio
Afghanistan	2020	0.0541
Afghanistan	2021	0.1011
Afghanistan	2022	0.097
Afghanistan	2023	0.0905
Albania	2020	0.0408
Albania	2021	0.0499
Albania	2022	0.0536
Albania	2023	0.0401
Nigeria	2020	0.1335
Nigeria	2021	0.1375
Nigeria	2022	0.1384
Nigeria	2023	0.1376
Angola	2020	0.0336

12.What is the energy consumption per capita for each country over the last decade?

Query

```
recent_years AS (  
  (year) AS max_year FROM consum_3),  
consumption_data AS (  
  country, c.year, c.consumption, p.value AS population_3 p  
  m_3 c  
  ation_3 p  
  ry = p.countries AND c.year = p.year  
  ar >= (SELECT max_year - 9 FROM recent_years)  
  untry, year,  
  consumption)/sum(population), 4) AS consumption_per_capita  
  mption_data  
  ountry, year  
  onsumption_per_capita desc;
```

Output

country	year	consumption_per_capita
Qatar	2023	0.0002
Bahrain	2020	0.0001
Kuwait	2020	0.0001
Qatar	2020	0.0001
Saudi Arabia	2020	0.0001
Singapore	2020	0.0001
Trinidad and Tobago	2020	0.0001
Bahrain	2021	0.0001
Kuwait	2021	0.0001
Qatar	2021	0.0001
Saudi Arabia	2021	0.0001
Singapore	2021	0.0001
Trinidad and Tobago	2021	0.0001
Bahrain	2022	0.0001

13.How does energy production per capita vary across countries?

Query

```
T
countries,
D(SUM(p1.production) / SUM(p.value
roduction_percapita
opulation_3 AS p
roduction_3 AS p1
.countries = p1.country
p.year = p1.year
BY p.countries
BY production_percapita DESC;
```

Output

countries	production_percapita
Qatar	0.0006
Kuwait	0.0003
Norway	0.0003
United Arab Emirates	0.0002
Australia	0.0001
Bahrain	0.0001
Kazakhstan	0.0001
Libya	0.0001
Mongolia	0.0001
Oman	0.0001
Saudi Arabia	0.0001
Trinidad and Tobago	0.0001
Turkmenistan	0.0001
Canada	0.0001

14. Which countries have the highest energy consumption relative to GDP?

Query

```
consumption)/sum(g.value),4) AS relative_consumption_for_gdp  
C  
  
= g.country  
g.year  
try  
relative_consumption_for_gdp desc;
```

Output

country	relative_consumption_for_gdp
Trinidad and Tobago	0.0051
North Korea	0.0045
Turkmenistan	0.0034
Bahrain	0.0022
Kuwait	0.0017
Iran	0.0017
Qatar	0.0015
Russia	0.0015
Kazakhstan	0.0014
Ukraine	0.0014
Iraq	0.0013
Canada	0.0012
United Arab Emirates	0.0012
Oman	0.0011

15.What are the top 10 countries by population and how do their emissions compare?

Query

```
SELECT
p.countries,
sum(p.value) AS total_population,
sum(e.emission) AS total_emission
FROM population_3 p
JOIN emission_3 e
ON p.countries = e.country
AND p.year = e.year
GROUP BY p.countries
ORDER BY total_population desc
limit 10;
```

Output

ountries	total_population	total_emission
na	22801232	92338
ia	22721260	20223
ted States	5334912.8	38453
onesia	4446374.8	5313
istan	3862739.1999999997	1640
eria	3534237.1999999997	874
zil	3358632.7999999993	3405
ngladesh	2699235.1999999997	894
ssia	2332911.5999999996	14481
xico	2051199.9999999998	3416

16.What is the global share (%) of emissions by country?

Query

```
country as (  
on) AS total_emission  
country)  
total_emission*100/(select sum(emission)from  
country
```

Output

Country	Share (%)
India	7.0850
Russia	5.0733
Japan	2.8507
Iran	2.0902
Indonesia	1.8614
Germany	1.8074
South Korea	1.7745
Saudi Arabia	1.7517

Conclusion

1. **Top Emitting Countries:** Nations like **China, the USA, and India** consistently rank highest in total emissions due to their large populations and industrial energy demands.
2. **High Emission Energy Types:** Fossil fuels (especially coal and oil) are the primary contributors to global emissions, highlighting the urgent need for cleaner alternatives.
3. **Per Capita Insights:** Some countries have high total emissions but low per capita emissions due to large populations. Others, mainly developed nations, show higher per capita emissions, though several have reduced theirs over the past decade.
4. **Energy Production vs. Consumption:** Countries like the USA and China show both high production and consumption, while others consume more than they produce, relying on imports.
5. **Energy Efficiency:** Countries with lower emission-to-GDP and consumption-to-GDP ratios demonstrate better energy efficiency, often due to cleaner energy policies and technologies.
6. **Global Emission Distribution:** A small number of countries account for a major share of global emissions, revealing uneven responsibility and opportunity for leadership in climate action.
7. **Population Impact:** Emissions generally rise with population growth, but countries that have invested in clean energy have successfully decoupled emissions from population increases.
8. **Global Averages:** Yearly averages show gradual changes, with global GDP and energy demand increasing, while emissions per capita show mixed trends depending on the region.

THANK
YOU

